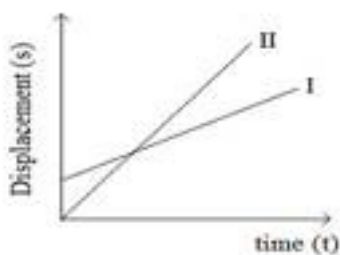


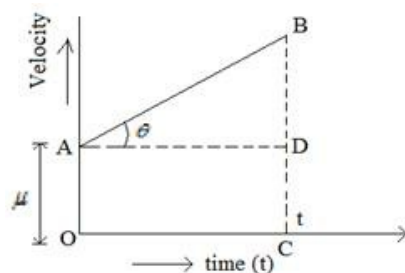
CBSE Test Paper 04
Chapter 3 Motion in A Straight Line

1. A ball starts from rest and accelerates at 0.500 m/s^2 while moving down an inclined plane 9.00 m long. When it reaches the bottom, the ball rolls up another plane, where, after moving 15.0 m , it comes to rest. What is the ball's speed in m/s 8.00 m along the second plane? **1**
 - a. 2.23 m/s
 - b. 2.56 m/s
 - c. 2.05 m/s
 - d. 2.15 m/s
 2. Path length is a **1**
 - a. tensor
 - b. Derived unit
 - c. scalar
 - d. vector
 3. While determining velocity from position-time graph. Velocity at any time is the slope of the _____ to the point on the graph at that instant. **1**
 - a. perpendicular
 - b. tangent
 - c. line from origin
 - d. Unit vector
 4. A bus start from rest with an acceleration of 1 m/sec^2 . A man who is 48 meter behind the bus with a uniform velocity of 10 m/sec . then the minimum time after which the man will catch the bus is **1**
 - a. 14 sec
 - b. 4 sec
 - c. 10 sec
 - d. 8 sec
-

5. A particle moves along the x axis. Its position is given by the equation $x = 2.00 + 3.00t - 4.00t^2$ with x in meters and t in seconds. Determine its position in m at the instant it changes direction **1**
- 3.21 m
 - 1.97 m
 - 2.22 m
 - 2.56 m
6. Can speed of an object be negative? Justify. **1**
7. What are positive and negative acceleration in straight line motion? **1**
8. A railway train 400m long is going from New Delhi railway station to Kanpur. Can we consider railway train as a point object. **1**
9. The position of an object moving along the x-axis is given by $x = a + bt^2$, where $a = 8.5$ m, $b = 2.5$ m and t is measured in seconds. What is its velocity at $t = 0$ s and $t = 2.0$ s? What is the average velocity between $t = 0$ s and $t = 4.0$ s? **2**
10. Sameer went on his bike from Delhi to Gurugram at a speed of 60km/hr and came back at a speed of 40km/hr. What is his average speed for entire journey? **2**
11. In Figure shows displacement - time curves I and II. What conclusions do you draw from these graphs? **2**



12. How does the velocity-time graph for uniform motion gives a geometrical way of calculating the displacement covered during a given time t? **3**
13. Establish $s = ut + \frac{1}{2}at^2$ from velocity time graph for a uniform accelerated motion? **3**



14. A woman starts from her home at 9.00 am, walks with a speed of 5 km h^{-1} on a straight road up to her office 2.5 km away, stays at the office up to 5.00 pm, and returns home by an auto with a speed of 25 km h^{-1} . Choose suitable scales and plot the x-t graph of her motion. **3**
15. On a two-lane road, car A is travelling with a speed of 36 km h^{-1} . Two cars B and C approach car A in opposite directions with a speed of 54 km h^{-1} each. At a certain instant, when the distance AB is equal to AC, both being 1 km, B decides to overtake A before C does. What minimum acceleration of car B is required to avoid an accident?